Cold Days, Warm Homes: The Impact of Winter Weather on Remote Work *

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Abstract

This study examines how winter weather conditions affect remote work patterns across U.S. metropolitan statistical areas. Combining individual-level telework data from the Current Population Survey with daily weather records from the National Oceanic and Atmospheric Administration, I employ a two-way fixed effects model and a continuous treatment difference in differences framework to estimate the causal impact of winter weather on labor supply decisions and trace the dynamic dose-response relationship between snow depth and telework hours. The empirical results demonstrate that adverse winter conditions, specifically lower temperatures and greater snow depth, significantly increase teleworking hours, with this effect markedly stronger in occupations characterized by high telework feasibility. Furthermore, while snowpack sharply reduces on-site work, telework partially compensates for the lost hours, mitigating the overall decline in total labor supply. The estimated dose-response curve confirms a robust, positive causal relationship between snow depth intensity and remote work substitution. These findings suggest that spatial reallocation of workplaces serves as an effective adaptation mechanism against weather-induced labor disruptions, offering important implications for climate adaptation policies and infrastructure planning.

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1 Introduction

Inclement winter weather conditions, including extreme cold and heavy snowfall, can severely disrupt economic activity. For instance, the blizzard that struck the Eastern United States in January 2015 imposed significant costs, particularly on the retail and transportation sectors. Severe winter conditions also impact labor markets, primarily by constraining labor supply: workers exposed to harsh weather often experience absenteeism and reduced working hours. Although global warming has gradually moderated cold extremes, researchers have found that the number of extreme regional snowstorms in the United States doubled between 1961 and 2010, relative to the period from 1900 to 1960.

Building on classic and frontier studies that link climate change to economic activity, recent research has increasingly examined how weather events shape individual behavior. Most of this literature has focused on heat waves and their economic consequences, such as losses in gross domestic product (García-León et al., 2021), lower labor productivity (Somanathan et al., 2021), and reduced working hours (Zivin and Neidell, 2014). Particularly, Zivin and Neidell (2014) found that workers adopt an adaptation strategy to extremely hot weather through temporal reallocation—reducing work hours and shifting time toward indoor leisure.

Despite mounting evidence that high temperatures impose significant economic losses, relatively little attention has been paid to the impact of winter weather on the economy, including its effects on the labor market. Bloesch and Gourio (2015) examined short-run effects of low temperatures and snow on employment and retail sales, finding modest aggregate impacts but sizable damages in specific sectors such as utilities, construction, and hospitality. Liu and Hirsch (2021) showed that snowfall reduces U.S. working hours, with heterogeneous impacts across workers, occupations, industries, and regions. Their results suggest that workers adapt to winter weather in a manner similar to that during heat waves, primarily through reduced working hours.

Yet, empirical evidence on alternative adaptation channels—beyond temporal reallocation—is still scarce. When workers must cut hours due to weather conditions, it is highly likely that they cannot fully offset lost income or utility costs, creating incentives to find alternative forms of adaptation that preserve working time. Spatial reallocation, which refers to the relocation of one's place of work from one location to another, could be a possible alternative. Furthermore, the spatial constraint of winter weather, driven by snowfall that causes reduced mobility and worsened outdoor work conditions, gives workers greater motivation to adopt strategies that relax location-based barriers to work, highlighting the need to study adaptation through spatial reallocation.

This paper proposes remote work as an adaptation strategy that mitigates the impact of adverse weather by reallocating work spatially, from on-site workplaces to homes or other preferred locations. In recent years, remote work has become a prominent alternative work arrangement, particularly following the COVID-19 pandemic. Before 2020, only 6.5 percent of U.S. private-sector employees primarily worked from home (Pabilonia and Redmond, 2024). The pandemic, however, served as a large-scale, unplanned experiment in telework, as public health concerns and lockdowns prompted millions of workers and firms to adopt remote work arrangements. Even after the pandemic, the prevalence of telework remained well above pre-pandemic levels in many sectors.

In this research, I estimate the impact of winter weather on telework hours as evidence of telework's role as an adaptive response to weather conditions. In parallel, I examine its effect on on-site hours to assess the extent of substitution between the two work modes. The analysis combines individual-level data on telework hours from the 2022–2025 Current Population Survey (CPS) with daily weather records from the National Oceanic and Atmospheric Administration (NOAA). Using a two-way fixed-effects (TWFE) estimation, I control for individual and temporal heterogeneity, and further estimate results by occupation-specific telework intensity as a proxy of telework feasibility of the occupation.

Beyond the baseline TWFE specification, I implement the continuous-treatment difference-in-differences framework of Callaway, Goodman-Bacon, and Sant'Anna (2024). This approach generalizes event-study designs to continuously varying treatments, allowing me to trace dynamic dose–response relationships between snow depth and telework hours and to reveal heterogeneous adaptation patterns beyond the average effects identified by the TWFE model.

This paper finds that lower temperatures and greater snow depth increase telework hours, while wind speed has negligible effects. Snow depth exerts the largest influence among occupations with higher telework feasibility, while the effects are statistically insignificant for jobs with lower telework intensity. Analyses using total and on-site hours confirm that additional snowfall reduces on-site work more strongly than total working hours, implying partial compensation of the loss in total working hours through telework. These findings suggest that telework effectively mitigates labor supply disruptions caused by winter weather.

Furthermore, the estimated dose-response curve suggests a similar conclusion to that of the TWFE estimation. Individuals who experienced a larger amount of snow during their survey period tend to have more hours of telework. The result not only aligns with the estimated results from the TWFE regression but also demonstrates that the positive relationship between snow and telework is robust, after integrating the facts into the empirical model that the treatment begins at different times (staggered) with varying amounts for individual workers (continuous).

This paper makes several contributions to the growing literature on weather adaptation and labor economics. First, it introduces telework as an emerging and effective behavioral adaptation strategy to adverse weather conditions, particularly winter weather that constrains mobility and outdoor activity. Second, it highlights the heterogeneity in telework feasibility across occupations as a key mechanism mediating workers' sensitivity to weather shocks. The increase in telework hours in response to cold or snowy

conditions is greater among occupations with higher telework feasibility, indicating that occupational characteristics play a significant role in shaping adaptive capacity. Third, the paper combines the results from both traditional two-way fixed effects models and a dynamic dose-response function with continuous treatment to robustly identify the causal effects of winter weather on telework behavior. Finally, the findings have broader implications for climate adaptation policy. Considering that various extreme weather events, such as floods or natural disasters, are likely to impose spatial constraints, promoting telework infrastructure could serve as a general adaptation measure to maintain economic activity in the face of diverse environmental shocks.

The remainder of the paper is organized as follows. Section 2 describes the data sources used in the analysis. Section 3 outlines the empirical strategy. Section 4 presents the main results on the impact of winter weather on telework hours. Section 5 concludes.

2 Data

2.1 Weather data

The weather data employed in this study are provided by the National Oceanic and Atmospheric Administration(NOAA). The data archived by National Centers for Environmental Information (NCEI) contains comprehensive and integrated historical weather datasets. One of the datasets is the Global Historical Climatology Network Daily (GHCN-Daily) dataset, which combines daily weather measurements from approximately 30 distinct sources, containing variables such as precipitation, daily high and low temperatures, wind speed, snowfall, and snow depth. These data are collected from over 90,000 terrestrial stations worldwide, with all temperature readings measured in Celsius. In this paper, I use the daily average temperature, wind speed, and snow depth to represent winter weather. The average temperature is the mean of the daily minimum and maximum temperatures. Since CPS provides weekly telework hours, the weather variables

are transformed to a weekly average.

The GHCN-Daily dataset contains two significant snow measurements: snowfall and snow depth. Snowfall measures the snow that fell in the last 24 hours, whereas snow depth measures the total snowpack on the ground, reported daily at 7 AM. This paper utilizes the snow depth instead of snow fall, to avoid the possible multicollinearity issue as the two variables are suspected to be highly correlated. Theoretically, the two variables are significantly different on the same day only if the snow fell for more than 24 hours. Drifting has little effect on the data because the observer records the average of multiple spots if snow drift is observed in most spots.

2.2 Telework hours data

The telework hours data used in this study are drawn from the IPUMS Current Population Survey (IPUMS-CPS), which harmonizes microdata from the monthly U.S. labor force survey. CPS respondents are asked to report information about events that occurred during the previous week (the "reference period"), specifically the week preceding the calendar week containing the 19th of the month. Accordingly, when integrating the CPS with weather data, I utilize daily temperature observations from the week that includes the 12th day of each month, aligning with the CPS reference period.

The CPS provides rich demographic information and is helpful for this study due to its records of how much the individual spent on remote work during the reference week. I focus on individuals residing in 252 Metropolitan Statistical Areas (MSAs), enabling a detailed examination of regional labor dynamics. Because the telework hours data is available from October 2022, the sample period of the dataset is from October 2022 to March 2025. Moreover, this paper limits the sample period from January to March and October to December each year to concentrate on the impact of winter weather.

A merit of using the CPS data lies in its longitudinal design, which enables the tracking of individuals during the same season over two successive years. This is made pos-

sible by the survey's unique structure: respondents are interviewed for four months in a row, take an eight-month break, and are then surveyed again for another four-month period. Consequently, the dataset provides repeated observations for individuals across two consecutive winters.

Table 1 presents summary statistics for the full sample and for selected occupational subgroups. Several patterns merit attention. First, the average telework hours vary substantially across occupational groups: workers in the 1st quartile report only 0.26 hours of telework per week, while those in the top decile report 18.81 hours, underscoring the wide disparity in telework feasibility across occupations. Correspondingly, the telework ratio rises from 0.01 in the 1st quartile to 0.51 in the top decile, indicating that workers in the most telework-intensive occupations spend roughly half of their working hours remotely.

Second, while total working hours are slightly higher for workers in the 4th quartile and top decile (37.85 and 37.44, respectively) compared to those in the 1st quartile (32.02), the composition of these hours differs markedly. Workers in the 1st quartile perform nearly all of their work on site (31.76 out of 32.02 hours), whereas those in the top decile split their hours roughly equally between telework (18.81) and on-site work (18.63). This compositional difference is central to the analysis, as it implies that the scope for substitution between on-site and remote work in response to weather shocks is substantially greater for higher-quartile occupations.

Third, weather conditions are largely similar across occupational groups, with average temperatures around 51°F and snow depth averaging 0.03 inches. This similarity is expected, as weather is determined by geographic location rather than occupation. The near-uniformity of weather exposure across groups helps alleviate concerns that the heterogeneous effects documented in the regression analysis are driven by differential exposure to winter weather rather than differential responsiveness.

Finally, the demographic composition of occupational groups reveals notable differ-

Table 1. Summary Statistics

	All (N = 625,048)		1st Quartile (N = 153,434)		4th Quartile (N = 180,469)		Top Decile (N = 65,868)	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Working hours								
Actual working hours	35.06	15.02	32.02	16.06	37.85	13.54	37.44	12.82
Telework hours	5.82	13.05	0.26	2.83	14.67	17.60	18.81	18.28
On-site working hours	29.24	17.85	31.76	16.17	23.18	19.32	18.63	18.66
Telework ratio	0.16	0.34	0.01	0.08	0.40	0.43	0.51	0.44
Weather conditions								
Average temperature (°F)	51.16	13.86	51.74	13.87	50.92	13.77	50.79	13.67
Snow depth (inches)	0.03	0.13	0.03	0.13	0.03	0.13	0.03	0.13
Snowfall (inches)	0.01	0.03	0.01	0.03	0.01	0.03	0.01	0.03
Average wind speed (miles per hour)	7.88	2.58	7.84	2.57	7.88	2.58	7.87	2.57
Regions and divisions								
New England	0.05	0.23	0.05	0.21	0.06	0.24	0.06	0.24
Middle Atlantic	0.08	0.27	0.08	0.27	0.08	0.27	0.08	0.27
East North Central	0.15	0.36	0.15	0.36	0.14	0.35	0.14	0.35
West North Central	0.05	0.22	0.05	0.21	0.05	0.22	0.05	0.23
South Atlantic	0.22	0.42	0.22	0.41	0.23	0.42	0.23	0.42
East South Central	0.05	0.21	0.05	0.21	0.04	0.20	0.04	0.19
West South Central	0.14	0.34	0.15	0.36	0.13	0.33	0.12	0.33
Mountain	0.08	0.27	0.08	0.27	0.08	0.27	0.08	0.27
Pacific	0.18	0.39	0.18	0.39	0.19	0.39	0.20	0.40
Demographics								
Age	42.02	14.62	38.98	15.48	44.06	13.36	42.21	12.84
Female	0.47	0.50	0.33	0.47	0.45	0.50	0.42	0.49
No HS diploma	0.08	0.27	0.20	0.40	0.01	0.10	0.00	0.06
HS graduate	0.40	0.49	0.60	0.49	0.21	0.41	0.18	0.38
Bachelor's degree	0.37	0.48	0.18	0.39	0.52	0.50	0.58	0.49
Graduate degree	0.16	0.37	0.02	0.13	0.26	0.44	0.24	0.43

Notes: Working hours are at the weekly level. On-site working hours equal the actual working hours minus telework hours. The telework ratio is measured as the share of telework hours in actual working hours for each worker; the number of observations for this variable is smaller than the full sample because the ratio is undefined when actual working hours equal zero. Weather conditions are the weekly average of daily observations; the number of observations for weather variables may differ from the reported sample size due to missing weather station data. The 1st quartile group includes workers in occupations with the lowest quartile of annual average telework hours across all occupations, while the 4th quartile group includes workers in occupations with the highest quartile. The top decile group comprises workers in occupations with the highest annual average telework hours.

ences. Workers in the 1st quartile are younger (38.98 years), more likely to be male (67 percent), and substantially less educated, with 20 percent holding no high school diploma and only 2 percent holding a graduate degree. In contrast, workers in the top decile are more educated, with 58 percent holding a bachelor's degree and 24 percent holding a graduate degree. These compositional differences further motivate the use of individual fixed effects in the baseline specification, which absorb time-invariant individual characteristics that may be correlated with both occupational choice and telework propensity.

3 Theoretical Framework

In this section, a simplified labor supply model is provided to analytically investigate the individual's decisions on teleworking. Since teleworking is a combination of spatial(where to work) and temporal(how many hours to work) choices, this paper suggests a model where the workers face a two-step decision problem regarding their work arrangement:

- **Step 1 (Extensive Margin):** The worker chooses *where* to work (on site or from home).
- **Step 2 (Intensive Margin):** The worker chooses *how many hours* to work.

Each worker solves the following utility maximization problem:

$$\max_{C,L} \; U(C, \tilde{L}) \tag{1}$$

subject to

$$C = w\left(H_s + \delta_k \cdot H_h\right), \tag{2}$$

$$L = T - H_s - H_h - \mathbf{1}(H_s > 0) \cdot T_c(W), \tag{3}$$

$$\tilde{L} = \varphi(W) \cdot L, \tag{4}$$

where C denotes consumption and L denotes leisure time. The worker allocates time between on-site working hours H_s, teleworking hours H_h, and leisure, subject to a total time endowment T. The utility function $U(C, \tilde{L})$ is strictly increasing and strictly concave in both arguments: $U_C > 0$, $U_{\tilde{L}} > 0$, $U_{CC} < 0$, and $U_{\tilde{L}\tilde{L}} < 0$.

Equation (2) is the budget constraint, where the worker earns a wage w per hour of on-site work. When teleworking, the worker's productivity is scaled by $\delta_k \in (0, 1)$, the relative telework efficiency for occupation k. This parameter represents how efficient

occupation k is when working from home compared to on-site work, so that the effective wage from teleworking is $w \cdot \delta_k$.

Equation (3) is the time constraint. The term $T_c(W)$ captures the temporal cost of commuting, which is increasing in winter weather severity W, i.e., $T_c'(W) > 0$. The indicator function $\mathbf{1}(H_s > 0)$ ensures that commuting costs are incurred only when the worker travels to the workplace.

Equation (4) introduces $\tilde{L}$, the *effective leisure*, which captures the idea that weather conditions directly affect the quality of leisure time. The function $\varphi(W)$ satisfies $\varphi(W) > 0$ and $\varphi'(W) < 0$: as winter weather becomes more severe, the enjoyment derived from a given amount of leisure time decreases. Note that we do not consider the direct impact of weather conditions on consumption.

Finally, the wage w, the winter weather severity W, and the telework efficiency δ_k are all treated as exogenous. For analytical tractability, this model excludes the option of "hybrid working," which would be a mix of on-site and telework. Therefore, the worker chooses one of the following:

1. **Commute and work at the workplace:** $H_s > 0$ and $H_h = 0$
2. **Work from home(telework):** $H_s = 0$ and $H_h > 0$

Case 1: Work at the workplace.

$$\max_{H_s} \; U(w \cdot H_s, \; \varphi(W) \cdot [T - H_s - T_c(W)]) \tag{5}$$

The first order condition is:

$$w \cdot U_C = \varphi(W) \cdot U_{\tilde{L}} \tag{6}$$

which implicitly defines the optimal on site working hours $H_s^*(w, W)$. The indirect utility

function of this case is:

$$V_{\text{site}}(w, W) = U(w \cdot H_s^*,\ \varphi(W) \cdot [T - H_s^* - T_c(W)]) \tag{7}$$

Case 2: Work from home.

$$\max_{H_h}\ U(w \cdot \delta_k \cdot H_h,\ \varphi(W) \cdot [T - H_h]) \tag{8}$$

The first order condition is:

$$w \cdot \delta_k \cdot U_C = \varphi(W) \cdot U_{\tilde{L}} \tag{9}$$

which implicitly defines $H_h^*(w, \delta_k, W)$. The indirect utility function is:

$$V_{\text{home}}(w, \delta_k, W) = U(w \cdot \delta_k \cdot H_h^*,\ \varphi(W) \cdot [T - H_h^*]) \tag{10}$$

3.1 Work Location Decision

Consequently, the worker decides to:

$$\text{Work at workplace if } V_{\text{site}} > V_{\text{home}}, \qquad \text{Work from home if } V_{\text{site}} < V_{\text{home}}.$$

Thus, the probability of working from home can be defined as:

$$P_{\text{WFH}} = P(V_{\text{home}} > V_{\text{site}}) \tag{11}$$

Without loss of generality, we can assume that P_{WFH} increases when V_{home} increases and/or V_{site} decreases.

4 Comparative Statics

4.1 Effect of Weather on the Probability of Teleworking

We apply the envelope theorem to each indirect utility function. Since H_s^* satisfies the first order condition (6), the effect of W on V_{site} is obtained by differentiating only through the explicit dependence on W, holding H_s^* fixed:

$$\frac{dV_{\text{site}}}{dW} = U_{\tilde{L}} \cdot \frac{\partial \tilde{L}}{\partial W}\bigg|_{H_s = H_s^*} = U_{\tilde{L}} \cdot \Big[\, \varphi'(W) \cdot L_s^* \; - \; \varphi(W) \cdot T_c'(W) \,\Big] \tag{12}$$

where $L_s^* = T - H_s^* - T_c(W) > 0$ is the optimal leisure time under on site work. Since $U_{\tilde{L}} > 0$, and both terms inside the brackets are negative—$\varphi'(W) < 0$ with $L_s^* > 0$, and $\varphi(W) > 0$ with $T_c'(W) > 0$—we have:

$$\frac{dV_{\text{site}}}{dW} < 0$$

This result reflects two distinct channels through which worsening weather reduces the value of on-site work:

1. **Commuting cost effect** $(-\varphi(W) \cdot T_c'(W))$: Severe weather increases commuting time, directly reducing the leisure time available to the on-site worker.

2. **Leisure depreciation effect** $(\varphi'(W) \cdot L_s^*)$: Severe weather reduces the quality of remaining leisure time.

Similarly, applying the envelope theorem to V_{home} using the first order condition (9):

$$\frac{dV_{\text{home}}}{dW} = U_{\tilde{L}} \cdot \varphi'(W) \cdot L_h^* < 0 \tag{13}$$

where $L_h^* = T - H_h^* > 0$. The home worker is also affected by weather through the leisure depreciation channel, but crucially, the commuting cost channel is absent.

Since both V_{site} and V_{home} decrease with worsening weather, the net effect on P_{WFH} depends on which decreases more. The key observation is that V_{site} is subject to *both* the commuting cost effect and the leisure depreciation effect, whereas V_{home} is subject *only* to the leisure depreciation effect. The additional negative term $-\varphi(W) \cdot T_c'(W)$ in equation (12) ensures that V_{site} declines by more than V_{home}, all else equal. Therefore:

$$\frac{dP_{\text{WFH}}}{dW} > 0 \tag{14}$$

Equation (14) establishes the extensive margin response to winter weather severity: as weather conditions worsen, workers are more likely to choose telework over on-site work, thereby increasing the probability of working from home.

4.2 Effect of Telework Efficiency on the Probability of Teleworking

Let $k = 1, 2, 3, \ldots, K$, and assume $0 < \delta_1 < \delta_2 < \cdots < \delta_K < 1$. A larger δ_k implies a greater telework efficiency. Workers in occupation m return more output when teleworking compared to workers in occupation $m - n$ $(0 < n \leq m - 1)$. The worker with occupation k earns an "effective wage" of $w \cdot \delta_k$ when working from home.

Applying the envelope theorem to V_{home} with respect to δ_k, using the first order condition (9):

$$\frac{dV_{\text{home}}}{d\delta_k} = \underset{(+)}{U_C} \cdot \underset{(+)}{w} \cdot H_h^* > 0 \tag{15}$$

Since V_{site} does not depend on δ_k, an increase in telework efficiency raises V_{home} while leaving V_{site} unchanged. Consequently:

$$\frac{dP_{\text{WFH}}}{d\delta_k} > 0 \tag{16}$$

That is, workers in occupations with higher telework efficiency are more likely to work

from home.

4.3 Effect of Weather on On-Site Working Hours

For workers who choose to work on site, we can express on-site working hours from the time constraint as:

$$H_s = T - T_c(W) - L \tag{17}$$

Totally differentiating with respect to W:

$$\frac{dH_s^*}{dW} = \underset{(+)}{-T_c'(W)} - \frac{dL^*}{dW} \tag{18}$$

The first term, $-T_c'(W) < 0$, is the commuting cost effect: worsening weather increases commuting time, which mechanically reduces the time available for on-site work, pushing H_s downward.

For the second term, note that as W increases, $\varphi(W)$ decreases, which lowers the marginal utility of leisure ($\varphi(W) \cdot U_{\tilde{L}}$). The worker optimally substitutes away from leisure toward work, so $dL^*/dW < 0$. This refers to the leisure depreciation effect: since bad weather makes leisure less enjoyable, the worker is motivated to work more, pushing H_s upward via $-dL^*/dW > 0$.

The two effects work in opposite directions. The commuting cost effect discourages on-site work, while the leisure depreciation effect encourages it. Without further parametric restrictions, the net effect of winter weather on on-site working hours is ambiguous:

$$\frac{dH_s^*}{dW} = \underbrace{-T_c'(W)}_{\substack{\text{commuting cost}\\ \text{effect } (<0)}} \underbrace{-\frac{dL^*}{dW}}_{\substack{\text{leisure depreciation}\\ \text{effect } (>0)}} \gtrless 0 \tag{19}$$

The sign depends on the relative magnitudes of the two effects, which is ultimately an empirical question.

5 Empirical Strategy

5.1 Baseline Model: Two-Way Fixed Effects Regression

To examine the effect of cold weather on remote work, this study adopts a two-way fixed effect (TWFE) model. This approach controls for both unobserved individual heterogeneity and temporal shocks. The proposed econometric specification is:

$$H_{ijt} = \beta_0 + \beta_1 T_{jt} + \beta_2 W_{jt} + \beta_3 S_{jt} + \alpha_i + \gamma_t + \epsilon_{ijt} \tag{20}$$

Where H_{it} denotes the total telework hours of individual i in a reference week of each month, living in the metropolitan statistical area (MSA) j at year-month t. The variable T_{jt} represents the weekly average of the daily average temperature in MSA j at time t, while W_{jt} and S_{jt} represent the weekly average wind speed and snow depth, respectively. Therefore, the estimated coefficients of β_1, β_2, and β_3 are expected to capture the impact of winter weather on telework hours.

The model includes the parameter γ_t to account for the time-fixed effect, specifically a year-month fixed effect for the analysis. This parameter is expected to capture the impact of macroeconomic conditions and weather seasonality on telework hours. The parameter α_i denotes the individual-fixed effect, capturing unobserved individual-specific characteristics such as skill levels, work effort. Another reason that individual fixed effects are crucial is that the tolerance level for cold weather differs for each individual, as well as their preference for remote work versus working at a site. The error term is denoted by ϵ_{it}, reflecting unobserved factors that influence individuals' labor supply decisions. Because individual fixed effects control for time-invariant individual characteristics, and time-fixed effects capture shocks common to all individuals at a given time, the remaining error term is assumed to reflect short-term unobserved variations across individuals, such as day-to-day fluctuations in physical condition, and is expected to be uncorrelated

with the independent variables.

5.2 Continuous-Treatment Event-Study Extension

To complement the baseline two-way fixed-effects (TWFE) specification, I further employ the continuous-treatment difference-in-differences framework developed by Callaway, Goodman-Bacon, and Sant'Anna (2024). This framework extends conventional event-study designs to settings in which both treatment timing and treatment intensity vary continuously across observational units. In the context of this study, the intensity of winter weather — measured by snow depth — is treated as a continuous treatment rather than a binary exposure. This approach enables the estimation of *dose-response functions* that track how telework hours evolve over time for the individuals experiencing varying levels of snow depth.

Formally, the framework identifies group-time-dose-specific average treatment effects,

$$ATT(g, t, d) = E[Y_t(g, d) - Y_t(0) \mid G = g, D = d], \tag{21}$$

where g denotes the first period in which an individual experiences a given level of snow, t denotes the observation period, and d represents the intensity of the treatment. By aggregating these effects over both event time and dose intervals, I obtain an event-study-type dose-response function,

$$ATT^{es}_{e_1,e_2}(d) = \frac{1}{e_2 - e_1 + 1} \sum_{e=e_1}^{e_2} E[ATT(G, G+e, d) \mid G + e_2 \in [2, T], D = d], \tag{22}$$

which summarizes the average response to a given level of treatment over a specific post-exposure window. The function $ATT^{es}_{e_1,e_2}(d)$ can be interpreted as a *dynamic dose-response curve*: for each level of treatment intensity d, it shows how the average telework response evolves across the two different posttreatment event times e_1 and e_2, such that $0 \leq e_1 \leq e_2$.

Callaway, Goodman-Bacon, and Sant'Anna (2024) exploit the two different event windows to compare the dose responses between the short-run and the long-run. However, to focus on the immediate response of the individuals to the snow and reduce the possibility of utilizing other adaptation strategies in the long run, I estimate the function with only one event window where $e_1 = 0$ and $e_2 = 1$. Therefore, in this research, the function $ATT^{es}_{e_1,e_2}(d)$ stands for the average treatment effect of being exposed to the level of snow depth d among all units that have been treated for at least 1 month, averaged over event-times 0 (the first month of the individual participated in CPS for given winter) to 1 (one month after the individual first participated in CPS for given winter). A positively sloped dose-response curve indicates that telework increases with the intensity of snow depth. Conversely, a flatter or negative slope would suggest limited adaptability or constraints on telework substitution under snowy weather.

Overall, this continuous-treatment DiD framework provides a flexible and informative characterization of how the intensity of snow depth causally affects telework behavior over time, revealing heterogeneous and dynamic adaptation patterns that would otherwise be obscured under standard TWFE models.

6 Results

Table 2 presents regression results estimating the effect of winter weather conditions on workers' telework hours. Panel A uses weekly average temperature as a continuous independent variable, while Panel B adopts temperature bins. Specifically, the low-temperature bin counts the number of days in each reference week with an average temperature below 20°F, the mid-temperature bin includes days between 20°F and 40°F, and the high-temperature bin covers days between 40°F and 60°F. Days above 60°F are omitted to avoid multicollinearity. Both panels additionally control for weekly average wind speed and snow depth.

Column (1) reports results for the full sample. In Panel A, a one-degree decrease in temperature is associated with a 0.015-hour increase in telework, while an additional inch of snow depth increases telework hours by 0.274. Panel B shows that all three temperature bins and snow depth are statistically significant, with colder days having a progressively stronger effect on telework hours.

Columns (2) through (5) report results by occupational quartiles, ranked by average telework hours. Column (2) corresponds to occupations in the lowest quartile, while Column (5) represents the highest. The fourth group, with the highest average telework hours, likely includes occupations more compatible with remote work, potentially implying greater receptiveness to telework. Comparing across Columns (2) to (5), I find that winter weather significantly affects telework hours only in occupations that are potentially receptive to telework. In contrast, telework hours among occupations in the lowest quartile, such as construction or mining, which are inherently unsuitable for telework, remain unaffected by winter conditions.

One notable pattern is the larger snow depth coefficients estimated in Column (4) for both panels. This may reflect the presence of occupations in the third quartile that are particularly vulnerable to snow. Column (6) isolates the top decile of occupations with the highest average telework hours. In Panel A, these occupations show strong responsiveness not only to cold and snow but also to wind, with the coefficient on snow depth increasing to 1.523 (Panel A) and 1.783 (Panel B). These results suggest that workers in highly telework-compatible occupations are especially sensitive to winter weather when making telework decisions.

When winter weather serves as an exogenous shock to work schedule decisions, workers may substitute telework for on-site work. However, for some workers, particularly those in outdoor occupations that are both vulnerable to winter weather and incompatible with remote work, both telework and on-site hours may decline. In such cases, winter weather results in a reduction of total working hours. Table 3 presents regression results

Table 2. The impact of winter weather on telework hours across occupational groups

	Telework hours					
	Full Sample (1)	1st Quartile (2)	2nd Quartile (3)	3rd Quartile (4)	4th Quartile (5)	Top 10 percent (6)
Panel A: Weekly average temperature						
Temperature	-0.015***	0.002	-0.019***	-0.014**	-0.025*	-0.040**
	(0.004)	(0.003)	(0.004)	(0.007)	(0.014)	(0.018)
Wind speed	-0.001	-0.008	0.010	-0.005	-0.009	-0.064
	(0.011)	(0.007)	(0.011)	(0.017)	(0.027)	(0.051)
Snow depth	0.274*	-0.036	-0.106	0.692***	0.636*	1.523**
	(0.145)	(0.097)	(0.153)	(0.264)	(0.364)	(0.601)
Panel B: Temperature bin						
Low-temperature bin (20°F or below)	0.091***	-0.012	0.156***	0.172***	0.078	0.091
	(0.035)	(0.021)	(0.039)	(0.061)	(0.094)	(0.141)
Mid-temperature bin (20°F to 40°F)	0.081***	-0.024*	0.099***	0.097**	0.118*	0.180*
	(0.024)	(0.013)	(0.024)	(0.040)	(0.064)	(0.101)
High-temperature bin (40°F to 60°F)	0.047***	-0.013	0.061***	0.064**	0.046	0.052
	(0.017)	(0.010)	(0.017)	(0.031)	(0.047)	(0.064)
Wind speed	-0.000	-0.009	0.010	-0.005	-0.006	-0.059
	(0.010)	(0.007)	(0.011)	(0.017)	(0.027)	(0.051)
Snow depth	0.367**	-0.068	-0.042	0.691**	0.812**	1.783***
	(0.152)	(0.098)	(0.159)	(0.284)	(0.373)	(0.591)
Observations	389,042	91,096	76,767	96,918	105,601	37,781

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table presents the estimated effects of winter weather on telework hours among workers in 252 metropolitan areas, using a two-way fixed effects model. The dependent variable is the number of telework hours. In Panel A, the independent variables include weekly averages of temperature (°F), wind speed (miles per hour), and snow depth (inches). Panel B replaces these continuous variables with three temperature bins. Column (1) shows results from the full sample, while Columns (2) through (5) present estimates for occupation-based quartile groups defined by average telework hours. Column (6) shows the results for the top decile of occupations with the highest average telework hours. The table reports estimated coefficients with robust standard errors in parentheses. All models include individual and time fixed effects. Robust standard errors are clustered at the MSA level.

using either actual working hours(Panel A) or on-site working hours(Panel B) as the dependent variable.

Column (1) shows that an additional inch of snow depth reduces weekly actual working hours by 0.313, while temperature has no statistically significant effect. In contrast, Panel B indicates that cold weather negatively affects on-site work hours, along with snow depth. These findings suggest that, on average, the *substitution effect*—where winter weather induces a shift from on-site work to telework—is stronger than the *output effect*, where winter conditions reduce both forms of work.

Table 3. The impact of winter weather on on-site working hours

	Full Sample (1)	1st Quartile (2)	2nd Quartile (3)	3rd Quartile (4)	4th Quartile (5)	Top 10 percent (6)
Panel A: Dep. var. = Actual working hours						
Temperature	0.002	0.020*	-0.002	-0.002	-0.002	0.008
	(0.007)	(0.011)	(0.013)	(0.009)	(0.009)	(0.014)
Wind speed	-0.060***	-0.067*	-0.069	-0.066**	-0.006	0.001
	(0.022)	(0.038)	(0.043)	(0.031)	(0.027)	(0.042)
Snow depth	-0.313*	-0.764*	-0.566	-0.261	0.036	0.648
	(0.159)	(0.435)	(0.360)	(0.350)	(0.264)	(0.423)
Panel B: Dep. var. = On-site working hours						
Temperature	0.017**	0.018	0.017	0.012	0.024*	0.048**
	(0.008)	(0.011)	(0.013)	(0.010)	(0.014)	(0.021)
Wind speed	-0.060**	-0.059	-0.079*	-0.061*	0.002	0.064
	(0.025)	(0.037)	(0.043)	(0.036)	(0.035)	(0.057)
Snow depth	-0.587***	-0.728	-0.459	-0.953**	-0.600*	-0.875
	(0.188)	(0.460)	(0.345)	(0.398)	(0.348)	(0.657)
Observations	389,042	91,096	76,767	96,918	105,601	37,781

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table presents the estimated effects of winter weather on actual and on-site working hours among workers in 252 metropolitan areas, using a two-way fixed effects model. In Panel A, the dependent variable is the total number of hours worked during the reference week, while the independent variables include weekly averages of temperature (°F), wind speed (miles per hour), and snow depth (inches). Panel B uses on-site working hours as the dependent variable, defined as the difference between total working hours and telework hours. Column (1) shows results from the full sample, while Columns (2) through (5) present estimates for occupation-based quartile groups defined by average telework hours. Column (6) shows the results for the top decile of occupations with the highest average telework hours. The table reports estimated coefficients with robust standard errors in parentheses. All models include individual and time fixed effects. Robust standard errors are clustered at the MSA level.

Notably, in Panel A, the temperature coefficient is positive and statistically significant only in Column (2), indicating that a one-degree decrease in temperature is associated with a reduction in total working hours for workers in the 1st quartile. For the remaining occupational groups, the temperature coefficients are small and statistically insignificant, suggesting that their total working hours are largely unaffected by cold weather. Combined with the positive effect of cold weather on telework hours documented in Table 2, this pattern implies that workers in telework-compatible occupations offset the loss in on-site hours through telework, thereby maintaining their total labor supply, whereas workers in occupations with low telework feasibility simply work fewer hours when temperatures fall.

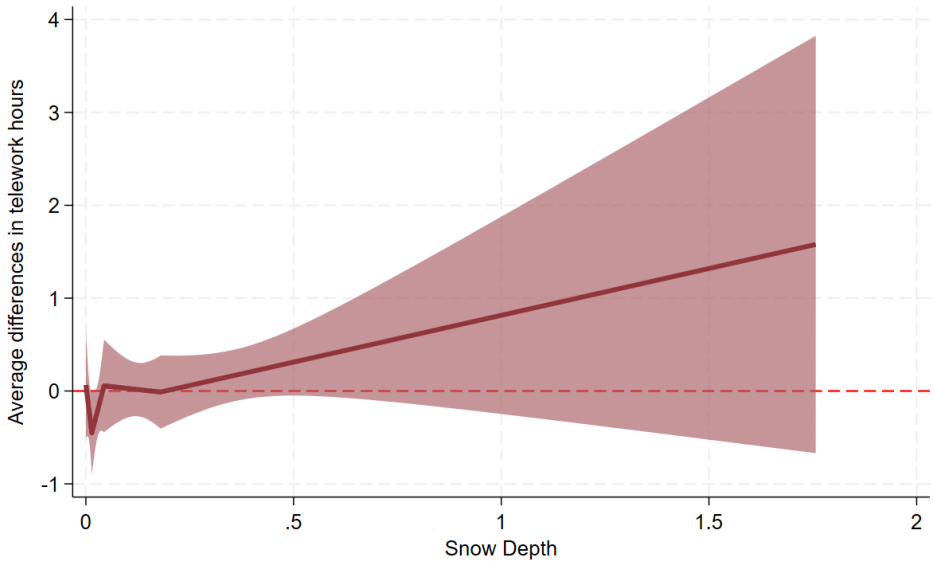

Figure 1. Estimated dose-response curve for the effects of snow depth

Notes: The figure shows the estimated dose-response function of telework hours with respect to snow depth intensity. Solid line denotes the estimate of $ATT^{es}_{e_1,e_2}(d)$ using Callaway, Goodman-Bacon, and Sant'Anna (2024). The shaded area is the 95% pointwise confidence intervals. The red line sets e_1 to 0 and e_2 to 1.

Figure 1 presents the estimated dose–response relationship between snow depth and telework hours based on the continuous-treatment difference-in-differences framework. The horizontal axis represents the average snow depth (in inches), while the vertical axis shows the estimated average effects on telework hours. The solid red line depicts the dose-response function using a linear spline with internal knots at the 25th, 50th, and 75th percentiles of the dose, and the shaded area indicates the 95% confidence interval.

The estimated dose–response curve shows a clear upward trend: as snow depth increases, telework hours tend to rise, suggesting that more severe snow conditions induce a greater shift toward remote work. Although the confidence band widens at higher snow depths—reflecting fewer observations of extremely snowy conditions—the estimates remain positive throughout, and the lower bound of the confidence interval is close to zero rather than negative. The estimation results from the quartile occupational groups with different telework feasibility are provided in the appendix.

While occupation-based categorization offers valuable insights, alternative classifica-

Table 4. The impact of winter weather on telework hours across divisions

	Telework hours								
	New England (1)	Middle Atlantic (2)	East North Central (3)	West North Central (4)	South Atlantic (5)	East South Central (6)	West South Central (7)	Mountain (8)	Pacific (9)
Temperature	0.030	-0.062**	-0.007	0.002	-0.050***	-0.033	-0.019	-0.009	-0.032*
	(0.071)	(0.025)	(0.020)	(0.027)	(0.014)	(0.024)	(0.012)	(0.011)	(0.017)
Wind speed	0.022	0.065	-0.055	0.079**	-0.054**	-0.102	0.035	0.020	0.006
	(0.030)	(0.042)	(0.056)	(0.029)	(0.024)	(0.060)	(0.029)	(0.049)	(0.020)
Snow depth	0.524	0.190	1.077***	0.540*	0.602	-1.191**	-1.362	-0.206	-0.800**
	(0.417)	(0.408)	(0.357)	(0.259)	(1.010)	(0.467)	(1.698)	(0.467)	(0.327)
Observations	25,781	18,276	60,386	35,777	74,121	25,156	60,475	47,628	41,442

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table presents the estimated effects of winter weather on telework hours among workers in 252 metropolitan areas, using a two-way fixed effects model. The dependent variable is the number of telework hours. The independent variables include weekly averages of temperature (°F), wind speed (miles per hour), and snow depth (inches). Each column presents results for one of the nine U.S. Census Bureau–designated divisions. The table reports estimated coefficients with robust standard errors in parentheses. All models include individual and time fixed effects. Robust standard errors are clustered at the MSA level.

tions may also yield statistically significant results and implications. One such approach is regional categorization, as geography is inherently linked to climate and weather conditions. Table 4 presents regression results by region, using the nine divisions defined by the U.S. Census Bureau to group observations geographically. If coefficient estimates are statistically significant and vary meaningfully across divisions, this could indicate that telework functions as a regional adaptation mechanism to winter weather (Liu and Hirsch, 2021).

However, statistical significance is observed only in selected divisions, such as East North Central and West North Central. Notably, the estimated effects for the South Atlantic and Pacific regions deviate from the earlier results in Table 2, suggesting opposite directional effects. Overall, the findings in Table 3 suggest that geographical location plays a limited role in shaping telework decisions in response to winter weather.

Table 5 presents estimation results by educational attainment, based on the hypothesis that higher education levels are positively associated with greater receptiveness to telework. While the temperature coefficients in Columns (4) and (5) are statistically significant, suggesting that telework hours among workers with a Bachelor's degree or higher

Table 5. The impact of winter weather on telework hours across education levels

	Telework hours				
	Full Sample (1)	No HS Diploma (2)	HS Graduate (3)	Bachelor's Degree (4)	Graduate Degree (5)
Temperature	-0.015***	0.003	-0.006*	-0.024***	-0.021*
	(0.004)	(0.007)	(0.003)	(0.008)	(0.011)
Wind speed	-0.001	0.002	-0.012	0.016	-0.016
	(0.011)	(0.014)	(0.010)	(0.017)	(0.045)
Snow depth	0.274*	0.357*	0.344	0.258	0.072
	(0.145)	(0.184)	(0.260)	(0.224)	(0.444)
Observations	389,042	26,293	151,400	144,931	60,606

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table presents the estimated effects of winter weather on telework hours among workers in 252 metropolitan areas, using a two-way fixed effects model. The dependent variable is the number of telework hours. The independent variables include weekly averages of temperature (°F), wind speed (miles per hour), and snow depth (inches). Column (1) shows results for the full sample, while Columns (2) through (5) present estimates for groups with different levels of education. All models include individual and time fixed effects. Robust standard errors are clustered at the MSA level.

increase in response to cold weather, the lack of consistent patterns across the education level indicates that education may not be a robust categorization criterion. Given the strong correlation between education and occupational choice, the statistically significant results in Table 4 are likely driven not by education itself, but by the concentration of telework-compatible occupations among more highly educated workers.

It is plausible to assume that age plays an important role in telework decisions. First, certain occupations, particularly those that are computer-based and information-intensive, are likely skewed toward specific age groups, especially prime-age workers. Second, age can be a critical barrier to adapting to winter weather, as older individuals are more susceptible to cold-related health risks and face a higher risk of falls due to snow accumulation.

Table 6 presents results by age group. The temperature coefficients are statistically significant for workers aged 35 to 64, with magnitudes ranging from -0.011 to -0.016, suggesting that cold weather increases telework hours primarily among middle-aged and older workers. This pattern is consistent with the concentration of telework-compatible occupations among prime-age and experienced workers who are more likely to hold positions with flexible work arrangements.

Table 6. The impact of winter weather on telework hours across age groups

	Telework hours						
	Full Sample	15–24	25–34	35–44	45–54	55–64	65+
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Temperature	-0.015***	-0.001	-0.004	-0.015**	-0.016**	-0.011*	-0.003
	(0.004)	(0.004)	(0.007)	(0.007)	(0.008)	(0.006)	(0.003)
Wind speed	-0.001	-0.007	0.026	-0.010	-0.011	-0.012	0.008
	(0.011)	(0.009)	(0.019)	(0.022)	(0.023)	(0.018)	(0.006)
Snow depth	0.274*	0.046	0.510*	0.271	-0.028	0.168	0.125
	(0.145)	(0.131)	(0.288)	(0.267)	(0.257)	(0.218)	(0.109)
Observations	389,042	88,792	94,768	102,523	95,516	98,212	160,150

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table presents the estimated effects of winter weather on telework hours among workers in 252 metropolitan areas, using a two-way fixed effects model. The dependent variable is the number of telework hours. The independent variables include weekly averages of temperature (°F), wind speed (miles per hour), and snow depth (inches). Column (1) shows results from the full sample, while Columns (2) through (7) present estimates for different age groups. All models include individual and time fixed effects. Robust standard errors are clustered at the MSA level.

The snow depth coefficient is positive and statistically significant only for the 25 to 34 age group (0.510). This may reflect the fact that younger workers in their early careers are more responsive to commuting disruptions caused by snow accumulation, potentially because they are more likely to live farther from their workplaces or to rely on transportation modes that are more vulnerable to snow. In contrast, snow depth does not significantly affect telework hours among older age groups, despite their greater susceptibility to cold-related health risks.

Taken together, the results suggest that the mechanisms driving weather-related telework decisions differ across age groups: for middle-aged and older workers, temperature is the primary driver, whereas for younger prime-age workers, snow-related commuting disruptions play a more prominent role. However, the evidence of heterogeneity across age groups is generally weaker and less consistent than that across occupational groups, limiting the interpretive strength in this dimension.

7 Robustness Checks

Table 7 replicates the baseline specification, replacing snow depth with snowfall as the measure of winter precipitation. While snow depth captures the accumulated snowpack on the ground, snowfall measures the amount of new snow that fell within the previous 24 hours. Although the two variables are closely related, they capture distinct dimensions of winter weather exposure: snow depth reflects persistent ground conditions that may affect commuting and mobility over multiple days, whereas snowfall captures the immediate disruption caused by an ongoing or recent snow event. Table 7 reports the estimation results using snowfall instead of snow depth, with all other specifications held constant.

The results are broadly consistent with the baseline findings. In Panel A, snowfall is positively and statistically significantly associated with telework hours for the full sample, with a coefficient of 1.612. Consistent with the baseline results, the effect is concentrated among occupations with higher telework feasibility: the coefficient increases from statistically insignificant estimates in the 1st and 2nd quartiles to 3.031 in the 3rd quartile, 3.338 in the 4th quartile, and 7.857 in the top decile. Panel B, which uses temperature bins, yields a similar pattern. These results confirm that the positive relationship between winter precipitation and telework is not sensitive to the choice of snow measurement, reinforcing the interpretation that snow-related disruptions drive spatial reallocation of work from on-site to remote settings.

Table 8 presents estimation results from a specification that excludes wind speed from the set of control variables. In the baseline results reported in Table 2, wind speed coefficients are statistically insignificant across most occupational groups, suggesting that wind conditions play a limited role in telework decisions. To verify that including wind speed does not distort the estimated effects of temperature and snow depth, I revisit the model without this variable.

The results remain broadly consistent with the baseline findings. In Panel A, the tem-

Table 7. Robustness Check: Replacing Snow Depth with Snowfall

	Telework hours					
	Full Sample (1)	1st Quartile (2)	2nd Quartile (3)	3rd Quartile (4)	4th Quartile (5)	Top 10 percent (6)
Panel A: Weekly average temperature						
Temperature	-0.014***	0.001	-0.015***	-0.013**	-0.026**	-0.040***
	(0.003)	(0.002)	(0.004)	(0.006)	(0.011)	(0.014)
Wind speed	0.004	-0.008	0.016	-0.003	-0.001	-0.061
	(0.010)	(0.006)	(0.011)	(0.015)	(0.023)	(0.042)
Snowfall	1.612***	-0.131	0.198	3.031**	3.338**	7.857***
	(0.618)	(0.393)	(0.607)	(1.325)	(1.679)	(2.960)
Panel B: Temperature bin						
Low-temperature bin (20°F or below)	0.066**	-0.008	0.124***	0.132**	0.062	0.049
	(0.029)	(0.018)	(0.034)	(0.056)	(0.078)	(0.126)
Mid-temperature bin (20°F to 40°F)	0.058***	-0.017	0.070***	0.051	0.098*	0.147*
	(0.021)	(0.011)	(0.022)	(0.036)	(0.056)	(0.082)
High-temperature bin (40°F to 60°F)	0.025	-0.011	0.046***	0.027	0.019	0.028
	(0.017)	(0.007)	(0.014)	(0.031)	(0.040)	(0.055)
Wind speed	0.004	-0.008	0.016	-0.004	-0.002	-0.061
	(0.010)	(0.006)	(0.010)	(0.016)	(0.023)	(0.043)
Snowfall	1.846***	-0.204	0.475	2.974**	3.580**	8.561***
	(0.650)	(0.364)	(0.629)	(1.328)	(1.789)	(3.170)
Observations	466,622	109,348	90,832	116,237	127,987	45,726

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table replicates the baseline specification from Table 2, replacing snow depth with snowfall (inches) as a measure of winter precipitation. The dependent variable is the number of telework hours. In Panel A, the independent variables include weekly averages of temperature (°F), wind speed (miles per hour), and snowfall (inches). Panel B replaces the continuous temperature variable with three temperature bins. Column (1) shows results from the full sample, while Columns (2) through (5) present estimates for occupation-based quartile groups defined by average telework hours. Column (6) shows the results for the top decile of occupations with the highest average telework hours. The table reports estimated coefficients with robust standard errors in parentheses. All models include individual and time fixed effects. Robust standard errors are clustered at the MSA level.

perature coefficient for the full sample is -0.014, comparable in magnitude and significance to the baseline estimate. The snow depth coefficients continue to exhibit the same pattern of increasing magnitude across occupational quartiles, rising from statistically insignificant estimates in the lower quartiles to 1.517 in the top decile. Panel B yields similar results, with snow depth coefficients that are positive and statistically significant for the 3rd quartile, 4th quartile, and top decile groups. The stability of these estimates confirms that the baseline results are not sensitive to the inclusion of wind speed as a control variable.

Table 8. Robustness Check: Excluding Wind Speed

	Telework hours					
	Full Sample (1)	1st Quartile (2)	2nd Quartile (3)	3rd Quartile (4)	4th Quartile (5)	Top 10 percent (6)
Panel A: Weekly average temperature						
Temperature	-0.014***	0.002	-0.018***	-0.012**	-0.024*	-0.039**
	(0.004)	(0.002)	(0.004)	(0.006)	(0.013)	(0.017)
Snow depth	0.283**	-0.013	-0.113	0.693***	0.675*	1.517***
	(0.141)	(0.093)	(0.147)	(0.252)	(0.355)	(0.573)
Observations	411,152	96,534	81,591	102,453	110,898	39,616
Panel B: Temperature bin						
Low-temperature bin (20°F or below)	0.035	-0.014	0.116***	0.140**	-0.037	0.005
	(0.039)	(0.020)	(0.042)	(0.056)	(0.102)	(0.133)
Mid-temperature bin (20°F to 40°F)	0.036	-0.025*	0.073***	0.067*	0.020	0.106
	(0.027)	(0.013)	(0.026)	(0.036)	(0.072)	(0.097)
High-temperature bin (40°F to 60°F)	0.022	-0.014	0.049***	0.045	-0.005	0.011
	(0.018)	(0.009)	(0.017)	(0.029)	(0.048)	(0.062)
Snow depth	0.461***	-0.033	0.063	0.563**	1.087***	2.018***
	(0.147)	(0.084)	(0.158)	(0.256)	(0.358)	(0.523)
Observations	437,675	101,905	86,397	108,483	119,919	42,874

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table replicates the baseline specification from Table 2, but excludes wind speed from the set of independent variables. The dependent variable is the number of telework hours. In Panel A, the independent variables include weekly averages of temperature (°F) and snow depth (inches). Panel B replaces the continuous temperature variable with three temperature bins. Column (1) shows results from the full sample, while Columns (2) through (5) present estimates for occupation-based quartile groups defined by average telework hours. Column (6) shows the results for the top decile of occupations with the highest average telework hours. The table reports estimated coefficients with robust standard errors in parentheses. All models include individual and time fixed effects. Robust standard errors are clustered at the MSA level.

It is worth noting that the sample sizes in this specification are larger than those in the baseline, as wind speed data are less available than temperature and snow depth records in the GHCN-Daily dataset. The fact that the estimated coefficients remain stable despite the change in sample composition further supports the robustness of the main findings.

A potential concern with the baseline specification is that some workers may change occupations during the sample period, potentially introducing variation in telework feasibility within individuals over time. Because the CPS follows respondents across two four-month interview periods separated by an eight-month gap, some individuals may have switched occupations between their first and second rotation. Such transitions could

Table 9. Robustness Check: Restricting to Occupation Stayers

	Telework hours					
	Full Sample (1)	1st Quartile (2)	2nd Quartile (3)	3rd Quartile (4)	4th Quartile (5)	Top 10 percent (6)
Temperature	-0.013**	-0.003	-0.011**	-0.011	-0.025	-0.037*
	(0.006)	(0.003)	(0.004)	(0.009)	(0.016)	(0.022)
Wind speed	0.000	-0.009	0.014	0.010	-0.013	-0.061
	(0.013)	(0.008)	(0.010)	(0.025)	(0.037)	(0.061)
Snow depth	0.337*	-0.183**	0.062	0.358	0.907*	1.532**
	(0.173)	(0.086)	(0.163)	(0.318)	(0.519)	(0.741)
Observations	223,049	48,775	53,688	56,075	64,511	20,843

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table replicates the baseline specification from Panel A of Table 2, but restricts the sample to workers who did not change occupations during the sample period. Because the CPS follows individuals across two four-month interview periods separated by an eight-month gap, some respondents may have changed occupations between their first and second rotation periods. Excluding these individuals ensures that each worker belongs to a single occupational quartile group throughout the sample, mitigating concerns about within-person shifts in telework feasibility. The dependent variable is the number of telework hours. The independent variables include weekly averages of temperature (°F), wind speed (miles per hour), and snow depth (inches). Column (1) shows results from the full sample of occupation stayers, while Columns (2) through (5) present estimates for occupation-based quartile groups defined by average telework hours. Column (6) shows the results for the top decile of occupations with the highest average telework hours. The table reports estimated coefficients with robust standard errors in parentheses. All models include individual and time fixed effects. Robust standard errors are clustered at the MSA level.

lead to abrupt changes in telework feasibility that confound the estimated weather effects, particularly in the quartile-based subgroup analyses, where a worker might shift from one occupational group to another.

To address this concern, Table 9 restricts the sample to workers who remained in the same occupation throughout their time in the sample. This restriction reduces the sample size from 389,042 to 223,049, a reduction of approximately 43 percent. Despite the substantially smaller sample, the results remain broadly consistent with the baseline findings. The temperature coefficient for the full sample is -0.013, comparable to the baseline estimate. The snow depth coefficients maintain the expected pattern of increasing magnitude across occupational quartiles, reaching 0.907 in the 4th quartile and 1.532 in the top decile, both of which are statistically significant. These results indicate that the baseline findings are not driven by individuals who experienced changes in occupational telework feasibility during the sample period.

Table 10 examines the sensitivity of the baseline results to alternative fixed effects structures. Panel A augments the baseline specification by replacing the year-month

fixed effects with state-by-year-month fixed effects, while retaining individual fixed effects. This more demanding specification absorbs state-specific time trends in telework behavior, ensuring that the estimated weather effects are identified solely from within-state, within-individual variation in weather conditions. The temperature coefficients remain negative and statistically significant for the full sample (-0.027), and the magnitudes are generally larger than the baseline estimates. The snow depth coefficient for the 3rd quartile is positive and statistically significant (1.505), consistent with the baseline pattern, although the estimates for the 4th quartile and the top decile lose statistical significance. This is likely because state-by-year-month fixed effects absorb much of the between-state variation in snow depth that the baseline specification exploits for identification, leaving only within-state, across-MSA differences in snow conditions, which tend to be small for geographically proximate areas.

Panel B adopts a different approach by replacing individual fixed effects with MSA fixed effects and additionally controlling for demographic characteristics, including age, gender, race, and education level. This specification is less restrictive than the baseline, as it does not control for unobserved individual heterogeneity, but it accommodates a larger sample and provides a useful benchmark. The larger sample size in Panel B reflects the fact that MSA fixed effects do not require repeated observations per individual, unlike individual fixed effects, which drop workers observed in only one survey rotation. The temperature coefficients are negative and statistically significant across most columns, with magnitudes increasing with higher telework intensity groups, reaching -0.063 in the top decile. However, the snow depth coefficients are largely insignificant, except for the 3rd quartile (0.546). The attenuation of the snow depth effect is mainly due to the absence of individual fixed effects, which, in the baseline specification, help isolate the weather response from unobserved individual characteristics correlated with the telework decision.

Table 11 presents results from a placebo test designed to assess whether the baseline

Table 10. Robustness Check: Alternative Fixed Effects Specifications

	Telework hours					
	Full Sample (1)	1st Quartile (2)	2nd Quartile (3)	3rd Quartile (4)	4th Quartile (5)	Top 10 percent (6)
Panel A: Individual FE + State × Year-Month FE						
Temperature	-0.027***	0.017*	-0.032***	-0.036*	-0.053*	-0.048
	(0.009)	(0.010)	(0.011)	(0.020)	(0.031)	(0.068)
Wind speed	-0.032*	-0.019	-0.004	-0.031	-0.078	-0.024
	(0.018)	(0.012)	(0.019)	(0.034)	(0.047)	(0.085)
Snow depth	0.561	-0.008	-0.290	1.505**	1.289	1.342
	(0.352)	(0.184)	(0.303)	(0.735)	(1.179)	(1.634)
Observations	389,042	91,096	76,767	96,918	105,601	37,777
Panel B: MSA FE + Year-Month FE + Demographic Controls						
Temperature	-0.018***	-0.002	-0.015***	-0.017*	-0.036**	-0.063***
	(0.005)	(0.002)	(0.005)	(0.009)	(0.014)	(0.020)
Wind speed	-0.013	-0.013**	-0.008	-0.012	-0.008	-0.031
	(0.014)	(0.006)	(0.015)	(0.025)	(0.038)	(0.063)
Snow depth	-0.010	0.037	-0.167	0.546**	-0.014	0.039
	(0.139)	(0.068)	(0.196)	(0.265)	(0.424)	(0.971)
Observations	434,973	108,674	90,669	114,623	121,007	43,967

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table presents estimation results under alternative fixed effects specifications. The dependent variable is the number of telework hours. Panel A replaces the baseline year-month fixed effects with state-by-year-month fixed effects while retaining individual fixed effects, thereby absorbing state-specific time trends in telework behavior. Panel B replaces individual fixed effects with MSA fixed effects and additionally controls for demographic characteristics including age, gender, race, and education level. The independent variables include weekly averages of temperature (°F), wind speed (miles per hour), and snow depth (inches). Column (1) shows results from the full sample, while Columns (2) through (5) present estimates for occupation-based quartile groups defined by average telework hours. Column (6) shows the results for the top decile of occupations with the highest average telework hours. The table reports estimated coefficients with robust standard errors in parentheses. Robust standard errors are clustered at the MSA level.

findings reflect a causal relationship between weather and telework, rather than spurious correlation. In this test, I replace the weather variables with their future values: Panel A uses weather conditions from one month after the CPS reference week, and Panel B uses conditions from two months after. If the baseline results capture a causal effect of weather on telework decisions, future weather, which workers cannot observe or anticipate at the time of their labor supply decisions, should have no systematic relationship with current telework hours.

The results are broadly consistent with this expectation. In Panel A, the temperature coefficients are small and statistically insignificant for the full sample and most sub-

groups, in contrast to the negative and significant estimates in the baseline specification. The snow depth coefficients are also largely insignificant, although a statistically significant estimate remains for the 4th quartile (0.753). This residual significance likely reflects the temporal autocorrelation of snow depth within a winter season. For instance, areas with deep snowpack in a given month tend to maintain similar conditions in the following month, generating a correlation between lead snow depth and current telework behavior. A few marginally significant coefficients appear in the top decile for temperature and wind speed, but their signs are positive, opposite to the baseline estimates, suggesting no meaningful relationship.

Panel B, which uses a two-month lead, provides a cleaner test, as the autocorrelation in weather conditions weakens over a longer horizon. The snow depth coefficients are uniformly small and statistically insignificant across all columns, with estimates close to zero or even negative, showing no resemblance to the large, positive baseline effects. While isolated marginal significance appears in a few cells, these are scattered across different variables and subgroups with no systematic pattern, consistent with what would be expected by chance at the 10 percent significance level. Consequently, the placebo results support the interpretation that the baseline findings reflect a contemporaneous causal response to weather conditions rather than an artifact of unobserved confounders or persistent seasonal patterns.

8 Conclusion

This study provides empirical evidence on how winter weather influences remote work behavior across U.S. metropolitan areas. Using individual-level telework data from the CPS combined with daily weather information from NOAA, the analysis demonstrates that adverse winter conditions - particularly low temperatures and increased snow depth - are correlated with greater telework hours, especially among occupations that are al-

Table 11. Placebo Test: Using Lead Weather Variables

	Telework hours					
	Full Sample (1)	1st Quartile (2)	2nd Quartile (3)	3rd Quartile (4)	4th Quartile (5)	Top 10 percent (6)
Panel A: One-Month Lead						
Temperature	0.005	0.003	0.004	0.001	0.010	0.026*
	(0.003)	(0.002)	(0.004)	(0.006)	(0.009)	(0.016)
Wind speed	0.006	0.003	0.022	0.006	0.005	0.102*
	(0.011)	(0.006)	(0.014)	(0.018)	(0.032)	(0.059)
Snow depth	0.141	0.175*	-0.089	-0.008	0.753**	0.892
	(0.116)	(0.090)	(0.124)	(0.165)	(0.307)	(0.555)
Observations	389,170	91,079	76,771	96,987	105,761	37,851
Panel B: Two-Month Lead						
Temperature	-0.000	-0.005**	0.004	0.008	-0.001	-0.003
	(0.003)	(0.002)	(0.004)	(0.006)	(0.009)	(0.017)
Wind speed	-0.001	-0.002	0.003	0.035*	-0.044	0.009
	(0.012)	(0.006)	(0.010)	(0.020)	(0.038)	(0.045)
Snow depth	0.003	-0.044	0.025	-0.048	0.309	-0.764
	(0.103)	(0.089)	(0.124)	(0.250)	(0.262)	(0.468)
Observations	388,744	90,953	76,680	96,947	105,697	37,814

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table presents results from a placebo test in which contemporaneous weather variables are replaced with future (lead) weather conditions. Panel A uses weather variables from one month after the CPS reference week, and Panel B uses weather variables from two months after. If the baseline results reflect a causal relationship between weather and telework, future weather conditions should have no predictive power for current telework hours. The dependent variable is the number of telework hours. The independent variables include weekly averages of temperature (°F), wind speed (miles per hour), and snow depth (inches), all measured at the specified lead. Column (1) shows results from the full sample, while Columns (2) through (5) present estimates for occupation-based quartile groups defined by average telework hours. Column (6) shows the results for the top decile of occupations with the highest average telework hours. The table reports estimated coefficients with robust standard errors in parentheses. All models include individual and time fixed effects. Robust standard errors are clustered at the MSA level.

ready telework-compatible. These findings suggest that remote work could successfully serve as an adaptive mechanism to winter weather, mitigating the disruption of labor supply during winter months for numerous occupations.

This paper examines heterogeneity in telework decisions across occupation, region, age, and education. The results indicate that workers in high telework-intensity occupations are significantly more responsive to winter weather conditions. While regressions by age and educational attainment yield some statistically significant estimates, the evidence of heterogeneity in these dimensions is generally weaker and less consistent than that observed across occupational groups, limiting the strength of interpretation in these

dimensions.

To policymakers, these findings highlight the value of remote work as an effective weather adaptation strategy. Encouraging telework arrangements can help sustain the labor supply during adverse weather events, particularly in groups prone to severe winters. Furthermore, to broaden the benefits of telework as a form of weather adaptation, this study offers guidance for targeted interventions by identifying specific occupations and demographic groups that should be prioritized.

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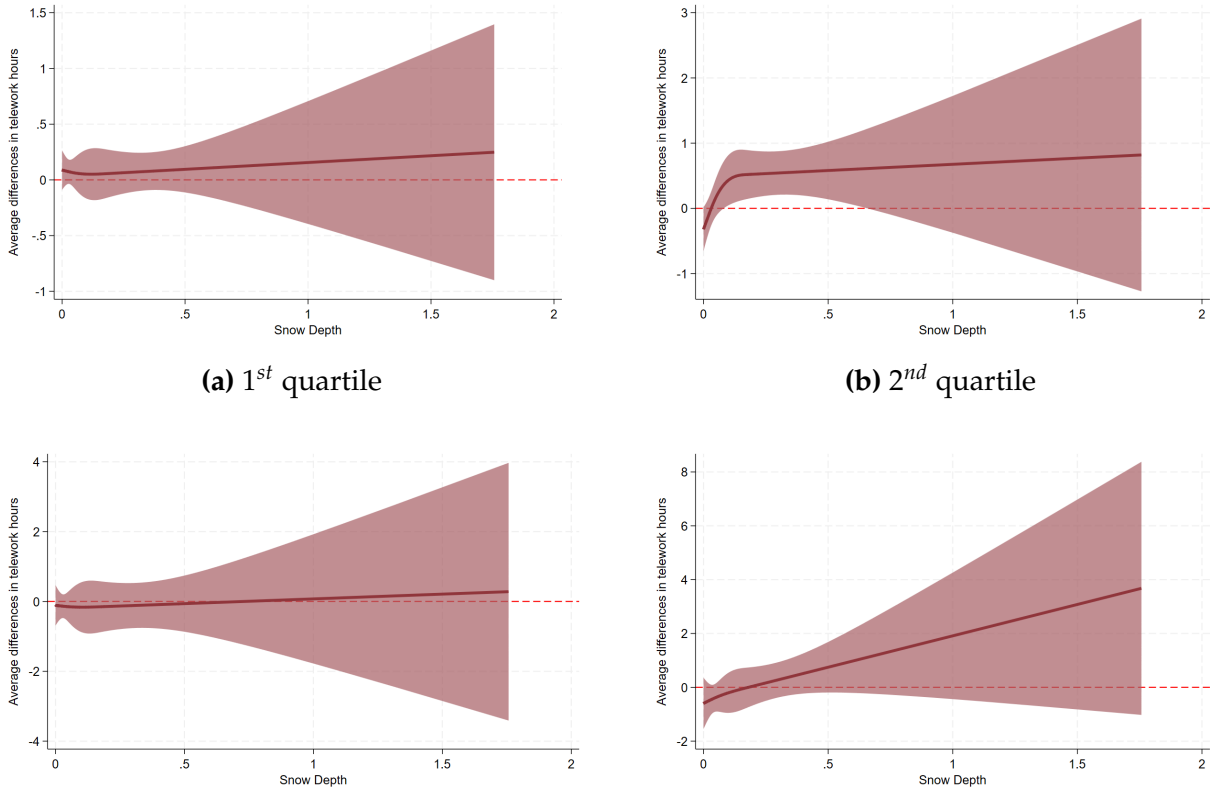
Appendix

Figure A1. Dose–response effects of snow depth on telework by occupational telework intensity



(a) 1st quartile

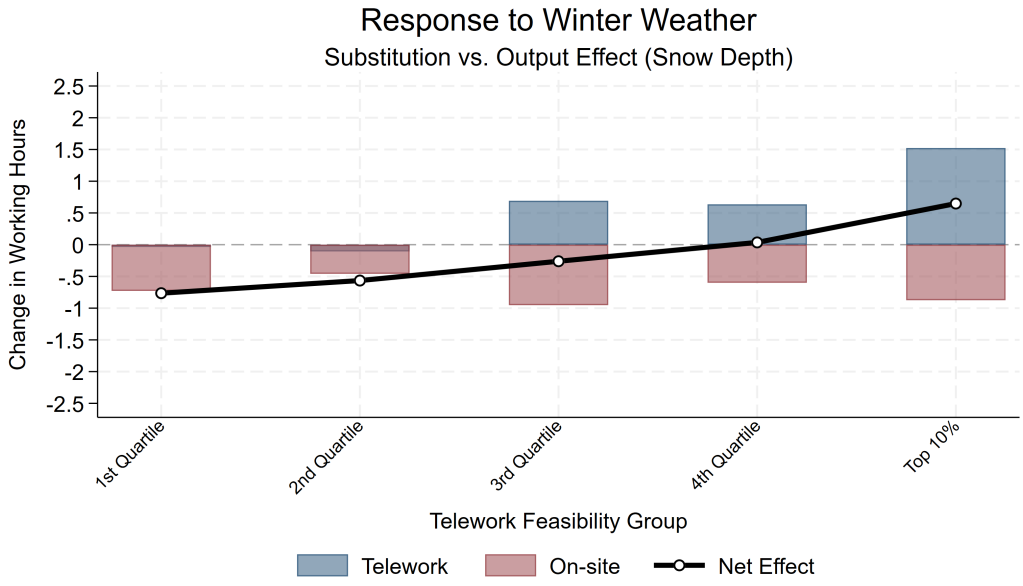
(b) 2nd quartile

(c) 3rd quartile

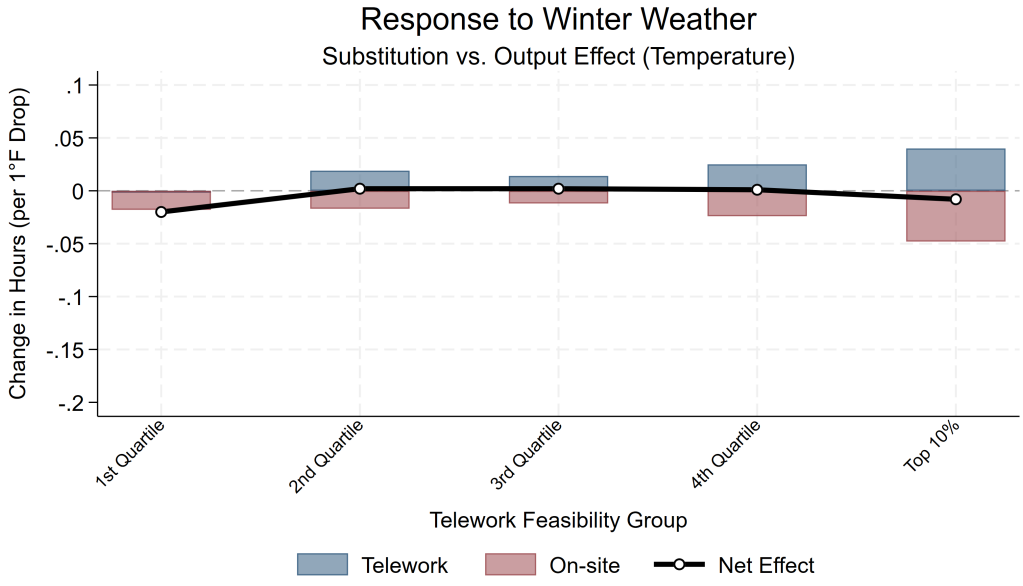
(d) 4th quartile

Notes: Each figure from (a) to (d) reports the estimated dose–response function between snow depth and telework hours for occupational groups with different levels of telework feasibility. Solid lines denote the estimate of $ATT^{es}_{e_1,e_2}(d)$ using Callaway, Goodman-Bacon, and Sant'Anna (2024). The shaded areas are the 95% pointwise confidence intervals. The red lines set e_1 to 0 and e_2 to 1. The figures suggest that the slopes are larger in the results with a higher level of telework intensity, which is consistent with the estimation results using the TWFE model.

Figure A2. Substitution vs. Output Effect of Winter Weather on Working Hours



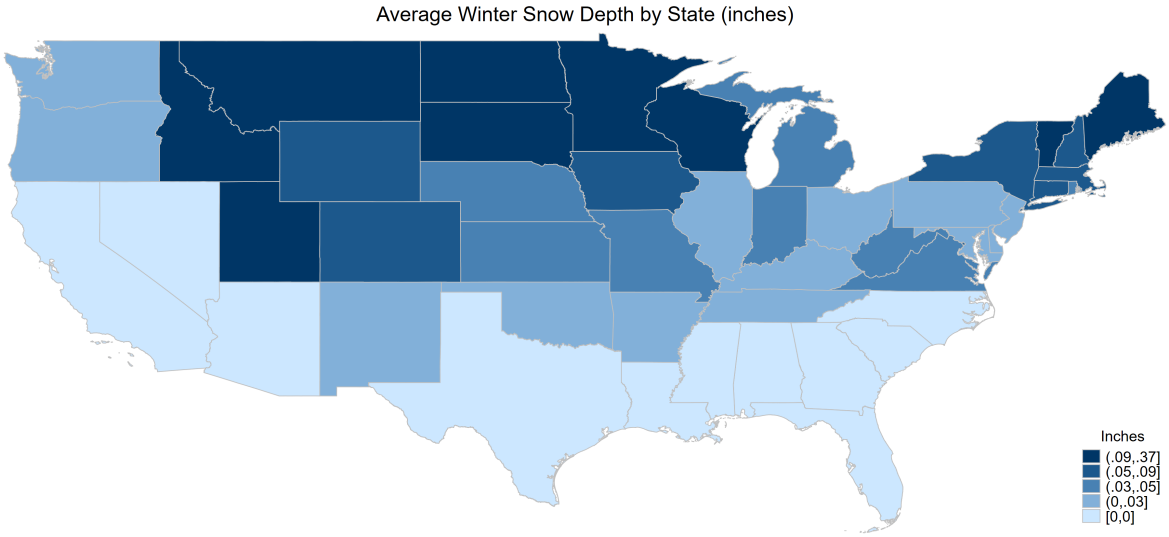
(a) Snow depth



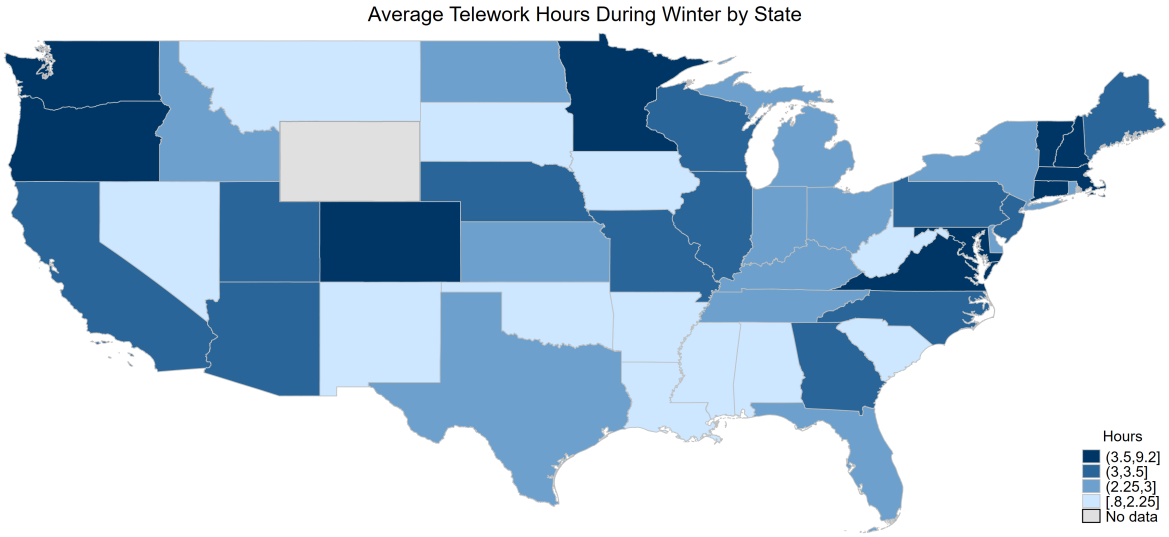
(b) Temperature

Notes: This figure visualizes the estimated effects of winter weather on telework hours (blue bars) and on-site working hours (red bars) across occupational groups, based on the regression results from Tables 2 and 3. Panel (a) shows the response to a one-inch increase in snow depth, and Panel (b) shows the response to a one-degree Fahrenheit decrease in temperature. The solid line with markers represents the net effect, defined as the sum of the telework and on-site hour changes, which corresponds to the change in total (actual) working hours. A net effect close to zero indicates that the substitution effect dominates, meaning that the reduction in on-site hours is largely offset by an increase in telework hours. A negative net effect indicates that the output effect dominates, meaning that winter weather reduces both telework and on-site hours, leading to an overall decline in total working hours. The pattern shows that the substitution effect becomes increasingly dominant as occupational telework feasibility rises, while the output effect prevails among occupations with low telework feasibility.

Figure A3. Winter Snow Depth and Telework Hours by State



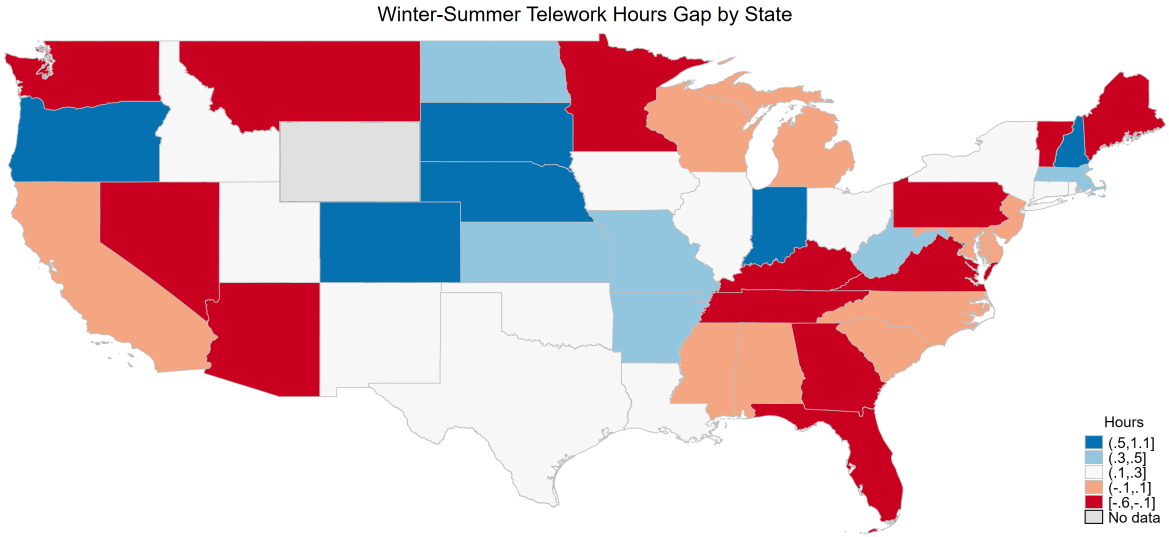
(a) Average winter snow depth



(b) Average winter telework hours

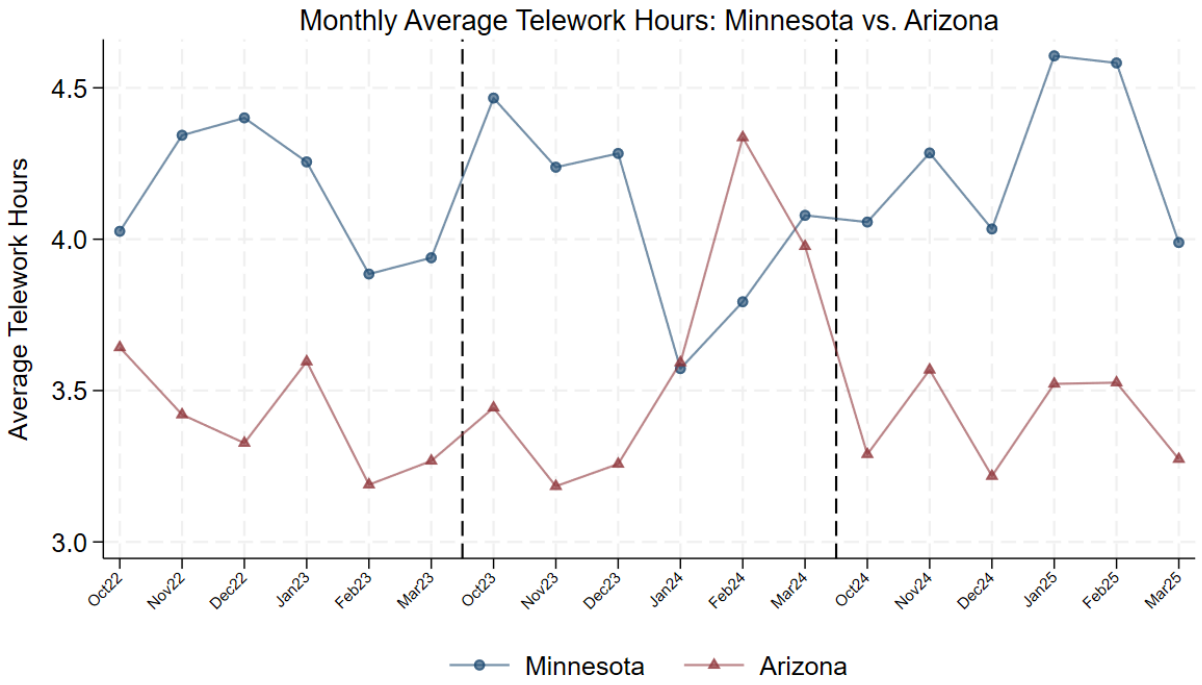
Notes: Panel (a) displays the average winter snow depth (in inches) by state, and Panel (b) displays the average winter telework hours by state. Both panels are based on the 2022–2025 sample period, with winter defined as October through March. Snow depth data are drawn from NOAA weather stations matched to 252 metropolitan statistical areas, and telework hours are drawn from the Current Population Survey. State-level averages are computed by aggregating MSA-level measures across all individuals within each state. States shown in gray indicate missing data. Alaska and Hawaii are excluded.

Figure A4. Winter-Summer Telework Hours Gap by State



Notes: This figure displays the difference between average winter and summer telework hours by state. Winter is defined as December and January, and summer is defined as June and July. Blue shading indicates states where telework hours are higher in winter than in summer (positive gap), while red shading indicates states where telework hours are lower in winter (negative gap). The data cover the 2022–2025 sample period, drawn from the Current Population Survey. State-level averages are computed by aggregating individual-level telework hours across all workers within each state. States shown in gray indicate missing data. Alaska and Hawaii are excluded.

Figure A5. Monthly Average Telework Hours: Minnesota vs. Arizona



Notes: This figure compares the monthly average telework hours between Minnesota (a cold-climate state) and Arizona (a warm-climate state) over the October 2022 to March 2025 sample period. Only winter months (October through March) are shown, with summer months excluded. Vertical dashed lines indicate boundaries between winter seasons. Telework hours are drawn from the Current Population Survey and averaged across all workers within each state for a given month. Minnesota consistently exhibits higher telework hours than Arizona, with the gap widening during the coldest months (December through February), suggesting a positive relationship between cold weather and telework behavior.

Table A1. Robustness Check: Finer Temperature Bins

	Telework hours					
	Full Sample (1)	1st Quartile (2)	2nd Quartile (3)	3rd Quartile (4)	4th Quartile (5)	Top 10 percent (6)
−20°F to −10°F	0.470*	-0.105	0.376**	0.078	1.070	0.988
	(0.281)	(0.088)	(0.189)	(0.281)	(0.752)	(1.042)
−10°F to 0°F	0.381***	-0.084	0.094	0.458*	0.951***	1.937***
	(0.132)	(0.086)	(0.080)	(0.235)	(0.272)	(0.634)
0°F to 10°F	0.041	0.055	0.251***	0.116	-0.120	-0.207
	(0.072)	(0.052)	(0.072)	(0.116)	(0.159)	(0.277)
10°F to 20°F	0.069*	-0.030	0.128***	0.152**	0.035	-0.036
	(0.038)	(0.031)	(0.039)	(0.068)	(0.101)	(0.172)
20°F to 30°F	0.082**	-0.017	0.099***	0.089*	0.144	0.318**
	(0.033)	(0.014)	(0.027)	(0.051)	(0.092)	(0.123)
30°F to 40°F	0.081***	-0.027*	0.100***	0.097**	0.118*	0.162
	(0.024)	(0.015)	(0.025)	(0.041)	(0.063)	(0.102)
40°F to 50°F	0.050**	-0.015	0.065***	0.059	0.068	0.109
	(0.020)	(0.012)	(0.021)	(0.036)	(0.054)	(0.084)
50°F to 60°F	0.043**	-0.011	0.056***	0.066**	0.031	0.030
	(0.018)	(0.009)	(0.016)	(0.032)	(0.048)	(0.062)
Wind speed	-0.001	-0.009	0.009	-0.005	-0.009	-0.065
	(0.010)	(0.007)	(0.011)	(0.017)	(0.027)	(0.052)
Snow depth	0.344**	-0.070	-0.036	0.696**	0.704*	1.494***
	(0.146)	(0.102)	(0.154)	(0.276)	(0.381)	(0.526)
Observations	389,042	91,096	76,767	96,918	105,601	37,781

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table replicates the temperature bin specification from Panel B of Table 2, but with finer bin widths of 10°F instead of 20°F. Each bin variable counts the number of days in the reference week with an average temperature falling within the specified range. The omitted category is days with temperatures above 60°F. The dependent variable is the number of telework hours. Column (1) shows results from the full sample, while Columns (2) through (5) present estimates for occupation-based quartile groups defined by average telework hours. Column (6) shows the results for the top decile of occupations with the highest average telework hours. The table reports estimated coefficients with robust standard errors in parentheses. All models include individual and time fixed effects. Robust standard errors are clustered at the MSA level.